Forecasting District Business Momentum (BRGI)

BRGI Datathon 2026 — Business Case Competition · Solution report · Team VAWAR

Final submission	Notebook	Result
solution_v6_w85.csv	brgi_solution_v6_w85.ipynb	Public leaderboard: **2.98849** (our best public score).

Executive summary

We forecast the Business Resilience & Growth Index (BRGI) of Indian districts 1–4 quarters after the 2024Q4 origin, giving a pessimistic (p10), central (p50) and optimistic (p90) value for each district and quarter. Our key insight is that BRGI is **exactly** a weighted sum of four cross-sectional ranks (35% credit growth, 20% deposit growth, 30% night-light growth, 15% credit resilience). So we built features that forecast each of those future ranks from the raw banking and satellite data, and fed them to gradient-boosted quantile models.

- Weighted pinball loss in rolling-origin backtests: **3.46**, against **4.7** for a naive "last BRGI ± 20" forecast (about 26% lower).
- Public leaderboard **2.988**. 2025 is calmer than the backtest years, which is why the leaderboard score is lower.
- Strictly point-in-time: we use only information published by the 2024Q4 origin, as defined by release_calendar.csv. No external data, no AutoML, and a fully reproducible notebook with fixed seeds.
- Districts with no banking history (about 15% of the test set) are forecast from night lights, census data and neighbouring districts. Their loss is only 6–23% higher than for districts with full history.

1. Problem and metric

Each test row is one district × target quarter (2025Q1–Q4, horizons h1–h4). The score is the mean pinball loss over the quantiles 0.1, 0.5 and 0.9, with horizons 3 and 4 weighted 1.25. Lower is better. We re-implemented this metric exactly and used it for every validation decision.

2. Data audit — what we verified

Check	Finding
Integrity	62,993 training rows and 2,472 test rows (618 rows per horizon). No duplicate IDs, no train/test ID overlap, no conflicting labels, and target_quarter = origin + horizon on every row.
Target structure	BRGI = 35·credit + 20·deposit + 30·night-light + 15·resilience score, verified to 1e-14. Every quarter has mean ≈ 50 and standard deviation 14–20 (rank-based).
Lag alignment	brgi_l1 is the label at origin − 1 (2024Q3 for the test set). The 2024Q4 labels appear only as training targets.
Publication lags	Bank data runs to 2024Q3 and night lights to 2024-11. Weather, fire and news data run to 2024-12-31.
Unseen districts	92 of the 620 test districts have no bank or BRGI history. They are a random-looking subset, spread across all regions.
Test-feature audit	We rebuilt every test column from the raw files. All of them match the information available at 2024Q4, so the test file contains no future information.
Data gaps	train.csv has no rows for origins 2024Q1–Q3 or 2016Q2–2017Q1, and bank data has a gap in 2016Q3–2017Q1.

3. Methodology

3.1 Point-in-time feature engineering

For a row with origin O and target T = O + h, every feature uses only the BRGI labels and bank data up to quarter O−1, and night lights up to the 2nd month of quarter O. Because BRGI is a sum of ranks, the most important features are *forecasts of the future ranks*, each computed across all districts at the same origin:

- **Credit and deposit growth (55%)**: the already-realised part of the year-on-year growth for quarter T (log level at O−1 minus log level at T−4), plus a projection of the remaining quarters at the recent growth rate.
- **Night-light growth (30%)**: a seasonal-level projection. We take the trailing 4-quarter log level, add the district's average seasonal deviation for the target calendar quarter, and compare that with the base quarter T−4. This raised the correlation with the realised rank from 0.25 to 0.38 at h1, and from about 0 to 0.25 at h4.
- **Resilience (15%)**: we found that the score is closely tracked by rank(mean − 0.5·sd of the last 4 quarterly credit growth rates), with correlation 0.83. We use this proxy for past quarters and for the part of the target window that is already known.
- **History and context**: BRGI lags (1–8 quarters), the same quarter last year, expanding mean and standard deviation, inverse-distance-weighted neighbour averages, and census/WorldPop static variables.

3.2 Models

- LightGBM quantile regression, with a separate model for each quantile (0.1, 0.5 and 0.9) and horizon used as a feature. Settings: num_leaves 15, learning rate 0.03, 600 rounds, bagging and feature subsampling. We wrote all the code ourselves and used no AutoML.
- **Two feature sets**: a *full* model for districts with history, and a *no-bank* model for unseen districts. The no-bank model uses night lights, census data and neighbour signals only. We checked automatically that no bank-derived feature enters it.
- **Blend**: an equal average of (a) a model trained directly on BRGI and (b) a model trained on the residual over the last BRGI (or over the neighbour mean for unseen districts). Each is averaged over 5 seeds.
- **Rebuilt recent origins**: train.csv has no rows for origins 2024Q1–Q3, so we rebuilt those rows from the raw files. Every rebuilt column was checked against existing origins. The weather, fire and news columns cannot be rebuilt, so they were dropped; in backtests they added nothing.
- **Calibration**: quantiles are sorted to remove crossings and clipped to [0, 100]. The p10/p90 distance from p50 is scaled by 0.85. Backtests on 2021–2023 favoured 1.1, but the 2025 leaderboard favoured narrower ranges.

4. Validation and results

Protocol. We used rolling-origin backtests at origins 2021Q4, 2022Q4 and 2023Q4, which share the test's Q4 seasonality. Each fold trains only on rows whose target is at or before the fold origin and predicts the next four quarters. In each fold, 15% of districts are removed from training entirely, to simulate the unseen test districts.

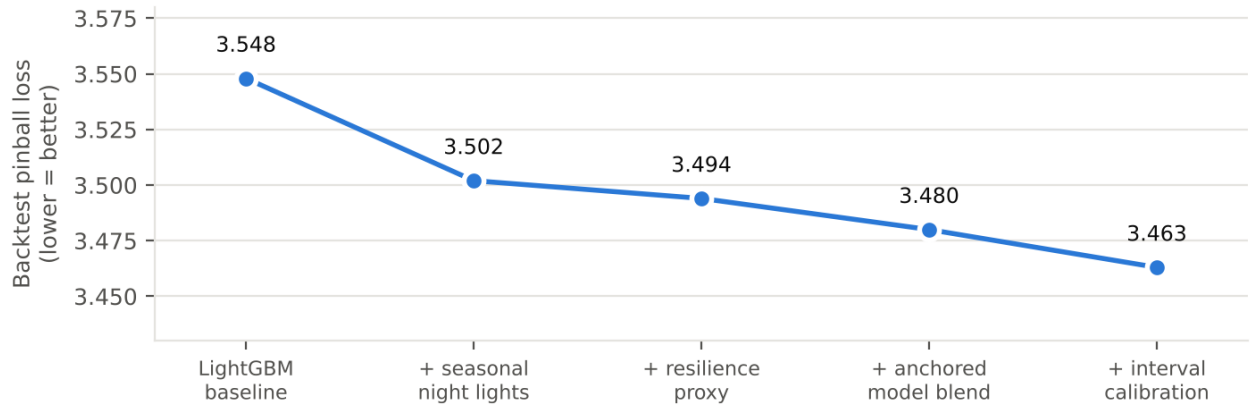


Figure 1. Mean backtest loss as each component was added. Each change was kept only when it improved the mean over the three folds.

4.1 Experiment log (main experiments)

Experiment	Backtest loss	Decision
Naive: last BRGI with ±20 band	≈ 4.70	Reference
LightGBM quantile baseline (given + engineered features)	3.548	Kept
+ seasonal night-light projections	3.502	Kept
+ resilience proxy features	3.494	Kept
+ blend with model anchored on last BRGI	3.480	Kept
+ interval calibration (×1.1 on backtest; ×0.85 chosen for 2025)	3.463	Kept
Rebuilt origins 2024Q1–Q3 (missing origins cost about 0.09; rebuilt rows score like real ones: 3.504 vs 3.503)	≈ –0.09	Kept
CatBoost multi-quantile	3.756	Rejected
Stacked component models (out-of-fold)	3.644	Rejected (leakage-prone)
State/region leave-one-out means	3.522	Rejected
Recency / Q4-origin / horizon sample weights	3.51	Rejected
Linear-tree LightGBM	3.801	Rejected
HDFC-merger credit adjustment	3.995 (2023Q4 fold)	Rejected

4.2 Leaderboard

Submission	Change	Public score
v3	Engineered features + resilience proxy, ranges ×1.1	3.00312
v5	+ rebuilt origins 2024Q1–Q3	≈ 2.990
v6_w85	Ranges ×0.85	2.98849

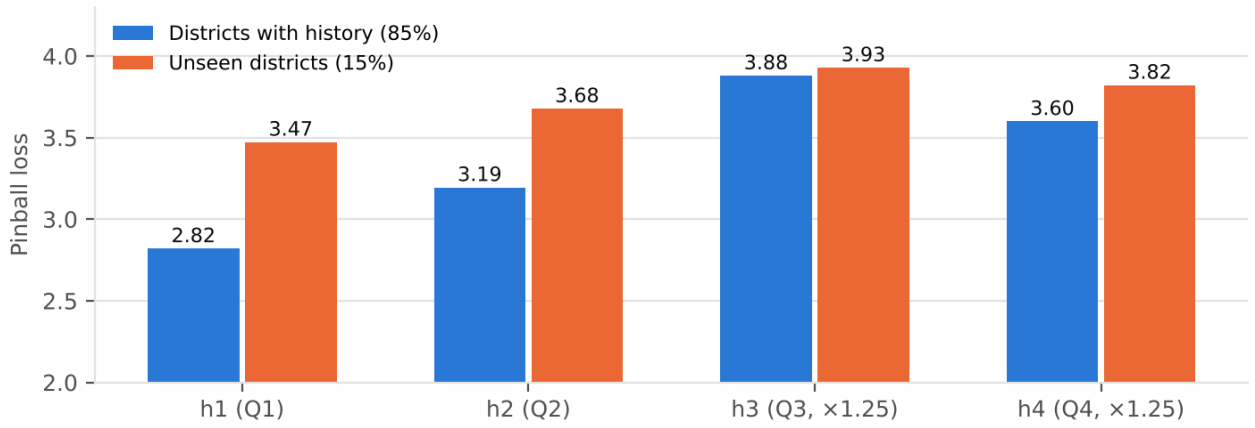


Figure 2. Backtest loss by horizon. Horizons 3–4 are the hardest and carry 1.25× weight. Unseen districts cost only 6–23% more thanks to night-light and neighbour features.

4.3 Where the remaining error comes from

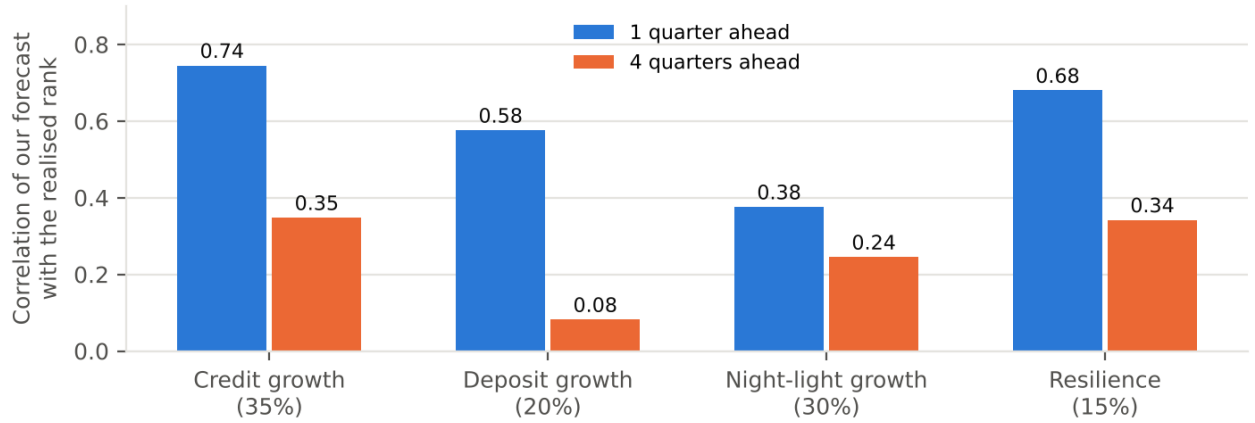


Figure 3. How well each BRGI component can be forecast. Credit growth is the most predictable. Deposit growth beyond two quarters is close to unpredictable from information available at the origin.

- The median absolute error of p50 is 8.9 BRGI points. The medians are well calibrated (45–52% of outcomes fall below p50 in every prediction quintile), and the backtest p10–p90 coverage is 78% against a nominal 80%.
- Training loss of 4.54 against validation loss of 5.39 for p50 shows moderate overfitting, not severe.
- Errors for the same district are only weakly persistent from year to year (correlation 0.22), so most remaining error is new information that appears after the origin.
- Northeast districts are hardest (loss 4.3, against 2.9 for Central), consistent with noisier night-light and banking series there.

5. Key findings

- **BRGI is a relative index.** A district's outlook depends on how fast its credit, deposits and night lights grow *relative to other districts*, not on its absolute size.
- **Credit momentum is the strongest lead indicator.** Growth already realised in the trailing year explains most of next quarter's credit score (correlation 0.75), and the signal decays steadily with horizon.
- **Night lights matter most where banking data is missing.** Seasonally adjusted night-light trends, combined with neighbours' momentum, give usable forecasts for districts with no banking history.
- **Uncertainty grows with horizon and is larger in the Northeast**, so planning decisions should use the p10–p90 range, not only p50.

6. Business recommendations

- **Credit and branch planning (banks, NBFCs)**: rank districts by forecast p50 BRGI for the next two quarters to prioritise branch expansion and lending campaigns. Districts with high p50 *and* a narrow p10–p90 range are the most dependable growth bets.
- **Early-warning watchlist**: flag districts whose p10 falls below 20 ("weak momentum") or whose forecast drops by more than 10 points from the last observed BRGI. These are candidates for portfolio-risk review and closer monitoring of collections.
- **Policy targeting**: use the forecasts for unseen or under-banked districts, which rely on night lights, to direct financial-inclusion programmes to places where economic activity is growing but banking has not yet followed.
- **Use the range, not just the point**: plan against p10 for capital and risk budgets and against p90 for capacity and upside, especially for 3–4-quarter decisions, where uncertainty is highest.
- **Refresh quarterly**: the pipeline is automated and point-in-time correct, so it can be re-run as each new RBI and Black Marble release arrives. The largest gains come from the newest banking quarter.

7. Limitations and next steps

- The weather, fire and news aggregates could not be rebuilt for the missing origins, so they were dropped. Rebuilding them from the raw daily files is a natural next step.
- Deposit growth beyond two quarters and night-light noise limit how accurate the forecasts can be. Better gap-filling of night lights could help.
- The interval width (×0.85) was chosen using leaderboard feedback on 2025. Backtests on earlier years preferred wider intervals, so the width should be re-checked each year.

8. Reproducibility and compliance

- Notebook **brgi_solution_v6_w85.ipynb** reproduces **solution_v6_w85.csv** exactly. We verified this as a byte-level match of all 2,472 rows after a clean end-to-end run. It runs in about 8 minutes on a Kaggle CPU and needs no internet.
- Python, pandas, numpy and LightGBM, with fixed seeds (42, 7, 2026, 11, 99) and deterministic LightGBM. No AutoML, no no-code tools and no cloud AI services.
- External datasets used: **none**. Districts are not de-anonymised. Every feature respects the publication lags in release_calendar.csv.